

A 2.2.1 Network Topologies



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Network topology details device connections, including star (central hub), mesh (interconnected), and hybrid types. Factors like cost, speed, and reliability influence choice. Star is good for homes, mesh for critical systems, and hybrids for large networks. Trends like SDN and cloud computing are shaping future flexible, scalable networks.

Introduction

A network topology defines how devices are physically or logically arranged and connected. Choosing the right topology involves trade-offs between cost, reliability, and performance, and different settings demand different solutions.

📌 **Exam focus (A2.2.1):** You need to know star, mesh, and hybrid topologies. Bus and ring are useful contexts, but are *not* assessed.

Topology comparison

Feature	Star	Mesh	Hybrid
Structure	All devices connect to a central hub or switch	Every device connects directly to every other device	Two or more topologies combined (e.g. star + mesh)
Reliability	Medium, hub failure takes down the network	High, multiple paths mean no single point of failure	High. inherits the strengths of component topologies
Transmission speed	Good, dedicated connection per device reduces congestion	Good, direct paths minimise latency	Varies by design; generally good for high-traffic zones

Feature	Star	Mesh	Hybrid
Scalability	Easy, add devices to the hub without disrupting others	Hard, each new device needs connections to all existing devices	Flexible, scale individual segments independently
Data collisions	Low, each device has its own link to the hub	Very low, dedicated paths per pair of devices	Low to moderate, depending on topology mix
Cost	Low, minimal cabling; hub/switch is inexpensive	High, $n(n-1)/2$ links required; complex to cable	Medium–high, depends on scale
Management	Simple, faults isolated to individual cable runs	Complex, large number of connections to maintain	Moderate, requires careful planning but is manageable by segment

Real-world contexts

Home / small office Star Simple to set up, low cost, easy to troubleshoot	Government / critical systems Mesh Downtime is unacceptable; redundancy is essential	Corporation/campus Hybrid (star + mesh) Different needs across departments; balances cost and resilience	Wireless mesh networks Mesh Extends coverage; nodes reroute traffic if one fails
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Key Questions

Can you answer these key questions relating to network topologies? Try to think and write your answers down after reading the text above or researching. Then you can click the eye icon to see some possible answers.

What are the advantages and disadvantages of each network topology?

Star is inexpensive and easy to manage but falls apart if the hub fails. Mesh is highly resilient and fast, but the cabling cost grows rapidly with each new device (n devices need $n(n-1)/2$ connections). Hybrid gains flexibility from combining topologies, but requires careful design to avoid complexity without benefit.

How does the choice of topology affect network performance and reliability?

Star topology reduces collisions by giving each device a dedicated channel to the switch, but reliability depends entirely on that central device. Mesh eliminates single points of failure and minimises latency through direct paths, making it ideal where uptime is critical. Hybrid designs allow engineers to apply mesh resilience only where it's needed, controlling cost.

What factors should be considered when choosing a network topology for a specific setting?

The five key factors from the syllabus are: **reliability** (tolerance for failure), **transmission speed** (bandwidth and latency requirements), **scalability** (ease of adding devices), **data collisions** (shared vs. dedicated links), and **cost** (cabling, hardware, and maintenance). These must be weighed against the specific setting and its criticality.

How do network topologies impact data security and privacy?

In a star network, all traffic passes through the central switch, making it straightforward to implement monitoring or a hardware firewall at that single point. In a mesh, multiple paths make traffic interception harder and complicate centralised monitoring. Hybrid networks can apply tight security to sensitive segments while keeping open access in public zones.

What are the emerging trends in network architecture, and how might they influence future network design?

Software-Defined Networking (SDN) separates the control plane from the data plane, allowing centralised, software-based management of complex hybrid topologies. Cloud computing shifts some infrastructure off-premises, reducing the need for large on-site mesh deployments. These trends increase flexibility and allow organisations to scale network resources dynamically.

Quiz

Take this quiz to test your knowledge of network topologies.

1

Which network topology connects all devices to a single central hub?

A. ☐ Mesh

B. ☒ Star

C. ☐ Bus

D. ☐ Ring

2

Which topology is MOST vulnerable to a single point of failure affecting the entire network?

A. ☒ Bus

B. ☐ Star

C. ☐ Mesh

D. ☐ Hybrid

3

Which topology offers the HIGHEST level of redundancy and fault tolerance?

A. ☒ Ring

B. ☐ Star

C. ☐ Bus

D. ☐ Tree

4

In which topology does data travel in a single direction around a closed loop?

A. ☐ Star

B. ☐ Tree

C. ☐ Bus

D. ☒ Ring

5

Which factor is LEAST important when choosing a network topology for a small home network?

A. ☒ Redundancy

B. ☐ Cost

C. ☐ Speed

D. ☐ Scalability

6

Which topology is MOST suitable for a large organization with multiple departments and buildings?

A. ☒ Hybrid

B. ☐ Star

C. ☐ Ring

D. ☐ Bus

7

What is a disadvantage of a mesh topology?

A. ☐ Slow transmission speed

B. ☒ High complexity

C. ☐ Limited scalability

D. ☐ Low reliability

8

Which topology is often used in wireless networks to extend coverage and improve signal strength?

A. ☐ Bus

B. ☐ Ring

C. ☐ Tree

D. ☒ Mesh

9

Which emerging trend allows for centralized control and management of network resources through software?

A. ☒ Software-Defined Networking (SDN)

B. ☐ Internet of Things (IoT)

C. ☐ Network Function Virtualization (NFV)

D. ☐ Cloud computing

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Which topology is MOST likely to experience data collisions?

A. ☐ Star

B. ☐ Tree

C. ☐ Mesh

D. ☐ Bus

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